30-Day Personalized Learning Roadmap

AI-Powered Career Guidance & Job Preparation System

Candidate Name:	Kriti Mahajan	**Candidate Rank:**	**Rank #1** of 1 (Readiness Score: **51.8%**)
Target Role:	**Python Backend Developer | AI Engineer**	**Current Skill Level:**	Beginner to Intermediate
Plan Scope:	30 Days (Structured 4-Phase Curriculum)	**Project Structure:**	**Dedicated 1-Day Hands-on Project** following each skill module

Gap Analysis Summary & Learning Strategy

Executive Summary: Based on the comprehensive gap analysis for the target role of Python Backend Developer | AI Engineer, the candidate demonstrates foundational competencies but requires targeted skill acceleration. Crucially, high-priority domain prerequisites currently categorized as Missing include Communication, Team Collaboration. To bridge these gaps swiftly and ensure candidate competitiveness, this 30-day personalized learning roadmap is engineered with high-impact, free practical learning resources. Each skill module is directly followed by a dedicated full-day practical project to ensure hands-on mastery before progressing to the next competency, culminating in an end-to-end multi-phase capstone portfolio project.

30-Day Learning Schedule

Day	Skill / Topic	Recommended Course	Optional Course
Phase 1 - Fundamentals & Core Skill Acquisition			
1	Linux: Command Line Mastery: Pipes, Redirection & Permissions	**Linux Command Line in 4 Hours** Click to open course	**linuxjourney.com** Click to open course
2	**[Project Day] Linux - Automated Bash Server Setup & Deployment Script**	**YouTube Educational Tutorial** Click to open course	**linuxjourney.com** Click to open course
3	FastAPI: Path Operations, Pydantic Models & Request Validation	**YouTube Educational Tutorial** Click to open course	**FastAPI Fast-Track Official Tutorial** Click to open course
4	**[Project Day] FastAPI - Production REST API with JWT Auth, Pydantic & PostgreSQL**	**YouTube Educational Tutorial** Click to open course	**fastapi.tiangolo.com** Click to open course
5	Docker: Dockerfiles, Image Building, Layers & Port Mapping	**YouTube Educational Tutorial** Click to open course	**Docker Hands-On Quickstart** Click to open course
6	**[Project Day] Docker - Multi-Stage Dockerfile & Containerized Backend Service**	**YouTube Educational Tutorial** Click to open course	**docs.docker.com** Click to open course
7	LangGraph: StateGraph Architecture, Nodes, Edges & State Schemas	**deeplearning.ai** Click to open course	**langchain-ai.github.io** Click to open course
Phase 2 - Intermediate Concepts & Practical Hands-on			
8	**[Project Day] LangGraph - Stateful Multi-Agent AI Workflow with Cyclical Execution**	**deeplearning.ai** Click to open course	**langchain-ai.github.io** Click to open course

Day	Skill / Topic	Recommended Course	Optional Course
9	Git: Branching Workflows, Staging, Merge Strategies & Conflicts	YouTube Educational Tutorial Click to open course	Git Fast Reference & Interactive Guide Click to open course
10	**[Project Day] Git - Production Multi-Branch Git Flow & Pull Request Template**	YouTube Educational Tutorial Click to open course	W3Schools Interactive Tutorial Click to open course
11	Communication: Principles of Data Storytelling & Visual Encoding	YouTube Educational Tutorial Click to open course	TCS iON Modular Course Click to open course
12	**[Project Day] Communication - Executive Data Storytelling Slide Deck & Brief**	YouTube Educational Tutorial Click to open course	TCS iON Modular Course Click to open course
13	Advanced RAG Optimization: Semantic Chunking Strategies, Metadata Enrichment & Embeddings	deeplearning.ai Click to open course	YouTube Educational Tutorial Click to open course
14	**[Project Day] Advanced RAG Optimization - High-Speed Hybrid Search Pipeline (Dense + BM25) with Re-Ranking**	deeplearning.ai Click to open course	YouTube Educational Tutorial Click to open course
Phase 3 - Advanced Problem Solving & Tool Mastery			
15	SQL: Relational Querying: SELECT, Filtering & Aggregations	YouTube Educational Tutorial Click to open course	SQL Interactive Exercises Click to open course
16	**[Project Day] SQL - E-Commerce Multi-Table Analytics Query Suite**	YouTube Educational Tutorial Click to open course	W3Schools Interactive Tutorial Click to open course
17	PostgreSQL: Schema Design, Constraints, Indexes & Normalization	YouTube Educational Tutorial Click to open course	PostgreSQL Essentials in 3 Days Click to open course
18	**[Project Day] PostgreSQL - Relational Database Schema Design with Foreign Keys & Indexes**	YouTube Educational Tutorial Click to open course	postgresqltutorial.com Click to open course
19	Python: Core Syntax, Data Structures & Modular Functions	YouTube Educational Tutorial Click to open course	Python Quickstart Tutorial Click to open course
20	**[Project Day] Python - Automated Data Cleaning & ETL Pipeline Script**	YouTube Educational Tutorial Click to open course	W3Schools Interactive Tutorial Click to open course
21	Team Collaboration: Code Review Culture, Pull Request Etiquette & Branch Policies	YouTube Educational Tutorial Click to open course	Team Collaboration Tutorial Click to open course
Phase 4 - Capstone Integration, Revision & Interview Readiness			
22	**[Project Day] Team Collaboration - Agile Sprint Planning, PR Code Review Checklist & Retrospective**	YouTube Educational Tutorial Click to open course	freeCodeCamp Tutorial Click to open course
23	Redis: In-Memory Data Structures, Key Expiration & TTL	YouTube Educational Tutorial Click to open course	Redis for Python Developers in 1 Day Click to open course
24	**[Project Day] Redis - High-Speed API Response Caching & TTL Expiration Layer**	YouTube Educational Tutorial Click to open course	redis.io Click to open course
25	AWS: Cloud Core: EC2 Compute, S3 Storage & VPC Networking	YouTube Educational Tutorial Click to open course	AWS Cloud Essentials Micro-Path Click to open course

Day	Skill / Topic	Recommended Course	Optional Course
26	**[Project Day] AWS - Deploy FastAPI App to AWS EC2 with RDS & S3 Storage**	**YouTube Educational Tutorial** Click to open course	**explore.skillbuilder.aws** Click to open course
27	Stakeholder Communication & Business Impact: Translating Technical Metrics to Business ROI & Stakeholder Alignment	**YouTube Educational Tutorial** Click to open course	**Executive Communication for Engineers** Click to open course
28	**[Project Day] Stakeholder Communication & Business Impact - STAR-Method Business Impact Brief on Tech Incident & Resolution**	**YouTube Educational Tutorial** Click to open course	**developers.google.com** Click to open course
29	**[Capstone Project (Phase 1)] End-to-End Multi-Tier Architecture & Database Schema Design**	**Capstone Implementation Guide** Click to open course	**Project Documentation & Reference** Click to open course
30	**[Capstone Project (Phase 2)] High-Concurrency Backend Microservice & AI Agent Integration**	**Capstone Implementation Guide** Click to open course	**Project Documentation & Reference** Click to open course

Important Guidelines: *(1) All recommended and optional courses are 100% free, publicly accessible resources. (2) Highlighted* ***Project Days*** *represent dedicated full-day practical implementations immediately following each skill module. (3) Click any course link to open the respective official learning platform directly.*